Fiscal Year:	FY 2025	**Task Last Updated:**	FY 05/22/2025
PI Name:	Perera, Imara Y Ph.D.		
Project Title:	Plant RNA Regulation Redux (in MVP--Multi-use Variable-g Platform)		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	None		
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Zip Code:	27695	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	Space Biology Unsolicited
Start Date:	05/31/2019	**End Date:**	12/31/2024
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:	Massa, Gioia	**Contact Phone:**	321-861-2938
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 12/31/2024 per NSSC (Ed., 6/13/24). NOTE: End date changed to 05/31/2024 per NSSC (Ed., 6/1/23). NOTE: End date changed to 05/30/2023 per NSSC (Ed., 5/27/22). NOTE: End date changed to 05/30/2022 per NSSC (Ed., 3/2/22).		
Key Personnel Changes/Previous PI:	Graduate Student Eric Land worked on this project until he completed his dissertation in April 2024; Aurora Toennisson designed the 3D holders for the gravitropism experiment and worked on data analysis. In addition, an undergrad student Kensi Bell assisted with characterizing genes of interest based on the transcriptomic analysis.		
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K1212		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The spaceflight experiment Plant RNA Regulation (PRR) was designed to generate comprehensive datasets for gene expression, small RNA species, and protein profiles from wild type Arabidopsis grown under the same set of physiologically relevant experimental conditions. Given the cost and limited availability of spaceflight opportunities, the goal was to maximize the scientific potential of the experiment. The experiment utilized the European Modular Cultivation System (EMCS) facility, which provides two critically important parameters for plant spaceflight experiments (Vandenbrink and Kiss, 2016). These are the capacity for providing a “1 g control in space” and a lighted environment for plant growth. PRR had two major objectives: To profile the transcriptional and post-transcriptional changes associated with early shoot and root development, and to monitor and characterize protein profiles of developing shoots and roots. Unfortunately, due to unknown factor(s), germination of PRR was reduced and growth of the seedlings was compromised. Based on poor growth and variability in growth, there was insufficient tissue to generate RNA for both RNA-Seq and small RNA sequencing for root and shoot tissue for all the samples. Additionally, there was insufficient tissue for the proteomic work. By careful examination of seedlings in the cassettes, we have been able to harvest individual seedlings from select seed cassettes that exhibited nominal growth and recover shoot and root tissue. A subset of these recovered shoot samples was processed using a modified RNA isolation protocol which enabled recovery of both RNA and sRNA species from the same sample. The RNA and sRNAs were subject to NextGen sequencing and preliminary analysis of these results has identified miRNAs differentially expressed between the 1g and micro g conditions. With the decommissioning of the European Modular Cultivation System (EMCS) it is not possible to have a reflight and repeat PRR in its original format. However we are confident that with some modifications and testing PRR can be adapted to fly in the newly developed Multi-use Variable-g Platform (Techshot) which would preserve two major criteria of PRR, namely the “1g control” in space and a lighted plant environment. This experiment will serve two important goals: This project has two major goals. 1. Accomplish/complete the science objectives of generating comprehensive datasets to compare plant responses to 1g vs micro g. 2. Provide a test and validation of the Phytofuge for plant experiments Reference: Vandenbrink J and Kiss JZ, 2016. Space, the final frontier: A critical review of recent experiments performed in microgravity. Plant Sci 243: 115-119.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This knowledge will provide a framework for designing and improving plants that are better adapted to adverse environments, which has potential benefit both for long duration space exploration as well as on Earth in the face of global climate change. In addition, this work will contribute towards enhancing education by providing ‘hands on’ training to undergraduate students in STEM (Science, Technology, Engineering and Mathematics) programs.
Task Progress:	We carried out a comprehensive time course (15, 30, 60, 90, 120, and 180 min after gravistimulation) of Arabidopsis root bending with the goal to identify early molecular events that take place in the root tip upon gravistimulation. We also included control (vertical) samples for each time point. Five replicate time course experiments were carried out and RNA isolated from 0.75 mm of the root tip. Samples were submitted for RNA-Sequencing, and the major results are summarized. 1. There was a significant transcriptional response at 30 min after bending (393 up-regulated and 167 down-regulated genes). Genes involved in ion transport, cellular signaling and stress responses were enriched in the up-regulated category with the plasma membrane as the enriched location. In contrast, genes down-regulated at 30 min were enriched in nuclear functions, histone methylation, and RNA and DNA modification. 2. Interestingly, there were few differentially expressed genes (DEGs) at the later time points compared to their matched controls. This shows that 15-30 min time period is important for transcriptional regulation. 3. In contrast, significant DEGs were identified at all of the gravistimulated time points, compared to the first control at 15 min. Additionally, comparison of the later control time points to the first control time point also revealed several DEGs. Further investigation shows that there is a strong time of day effect with several light and clock regulated genes showing a distinct pattern of regulation across the time course irrespective of the treatment. 4. These results reveal the dynamic and cycling nature of a majority of genes and highlight the importance of having time-matched controls for time course experiments.
Bibliography Type:	Description: (Last Updated: 05/30/2025)
Abstracts for Journals and Proceedings	Land ES, Toennisson A, Perera IY. "Early transcriptional response of gravistimulated Arabidopsis root tips." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Poster Session. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024